FINAL EXPLANATION: MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: MATHEMATICS GRADE 10 — SUB-STRAND 3.4: LINEAR MOTION	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation task. You have spent 10 lessons exploring how mathematics helps us describe, graph, and predict motion in a straight line. Now it is time to bring everything together into one well-reasoned, evidence-based explanation.

Your goal is to answer the driving question fully and clearly:
"How can mathematics help us fully describe, graph, and predict the motion of objects moving in a straight line — and why does direction always matter?"

Think back to the Timer Race Puzzle at the start of the unit. Three walkers left the same point, covered similar distances, and arrived (or ended up) at very different locations. That puzzle is the thread that connects every section below. In each section, use mathematical language, calculations, and graphs to show that you truly understand — not just what the answer is, but WHY it works the way it does.

READ ALL INSTRUCTIONS BEFORE YOU BEGIN:

- Answer every section in full sentences unless a diagram or calculation is specifically requested.
- Show all working for every calculation — unsupported answers will not earn full marks.
- Use correct units at all times (metres, seconds, m/s, m/s², etc.).
- You may refer to your notes, graphs, and any work completed during the unit.
- Where a graph is required, label all axes with quantities AND units, plot at least three clearly marked points, and draw the line or curve accurately.
- Connect your answers back to the Timer Race Puzzle and/or to real Kenyan contexts (e.g., a matatu on Thika Road, Eliud Kipchoge's marathon, a boda boda in Kisumu) wherever you can — this shows deep understanding.
- Write in full mathematical sentences: define every symbol you use the first time it appears.

Section 1: Scalar vs. Vector — Why Direction Changes Everything	
Explain the difference between scalar quantities (distance and speed) and vector quantities	Distance is a scalar quantity — it measures the total length of the path travelled, regardless of direction. Displacement is a vector quantity — it measures the shortest straight-line distance from start to finish AND states the direction. Speed is a scalar (how fast, no direction), while velocity is a vector (how fast AND in which direction).

(displacement and velocity). In your explanation: **(a) Define all four quantities clearly in your own words, and state whether each is scalar or vector.** **(b) A learner walks 400 m north along Kenyatta Avenue, then turns and walks 300 m east. Calculate:** **(i) the total distance travelled** **(ii) the magnitude and direction of the displacement from start to finish (use Pythagoras' theorem and trigonometry — show full working).** **(c) Using the Timer Race Puzzle as your example, explain why two walkers who cover the same distance can end up at completely different locations. Your answer must use the words 'scalar' and 'vector' correctly.**	(b)(i) Total distance = 400 m + 300 m = 700 m. (b)(ii) Because the two parts of the journey are at right angles, I can use Pythagoras' theorem: $\|\text{displacement}\| = \sqrt{(400^2 + 300^2)} = \sqrt{(160\,000 + 90\,000)} = \sqrt{250\,000} = 500$ m. For direction, using trigonometry: $\tan \theta = 300/400 = 0.75$, so $\theta = \tan^{-1}(0.75) \approx 36.87° \approx 36.9°$ east of north. Displacement = 500 m at 36.9° east of north. (c) In the Timer Race Puzzle, the zigzag walker and the straight-line walker both covered the same scalar distance, so by a scalar measure they are 'equal'. But distance is a scalar — it ignores direction entirely. Because the zigzag walker changed direction multiple times, their displacement (the vector from start to finish) was much smaller than the straight-line walker's. The straight-line walker's displacement equalled their distance, because they never changed direction. This is exactly why a matatu's speedometer tells you how fast it is moving but cannot tell you whether it is heading towards Nairobi or away from it — speed is scalar, velocity is vector.

Section 2: Calculating Speed, Velocity, and Acceleration	
Use correct formulae to solve the following problems, showing all working and units. **(a) Write down the formulae for:** **(i) average speed** **(ii) average velocity** **(iii) acceleration** **For each formula, define every symbol and state the SI unit of the result.**	(a)(i) Average speed = total distance ÷ total time. Symbol: $\bar{v}$ (or $\bar{s}$); unit: metres per second (m/s). It is scalar. (a)(ii) Average velocity = total displacement ÷ total time. Symbol: $\vec{\bar{v}}$; unit: m/s. It is vector — direction must be stated. (a)(iii) Acceleration = change in velocity ÷ time taken = $(v - u) \div t$. Symbols: a = acceleration (m/s^2), v = final velocity (m/s), u = initial velocity (m/s), t = time (s). (b) Let west be negative and east be positive. Displacement west: −1200 m; displacement east: +800 m. (i) Total distance = 1200 + 800 = 2000 m. (ii) Total displacement = −1200 + 800 = −400 m, i.e. 400 m due west. (iii) Total time = 3 + 1 + 2 = 6 min = 360 s. Average speed = 2000 ÷ 360 ≈ 5.56 m/s. (iv) Average velocity = −400 ÷ 360 ≈ −1.11 m/s, i.e. 1.11 m/s due west.

(b) A boda boda rider in Kisumu travels 1.2 km due west in 3 minutes, stops for 1 minute, then travels 0.8 km due east in 2 minutes. Calculate: **(i) the total distance travelled** **(ii) the total displacement (magnitude and direction)** **(iii) the average speed for the whole journey** **(iv) the average velocity for the whole journey.** **(c) A matatu on Thika Road accelerates from rest to 25 m/s in 10 seconds, then brakes uniformly to a stop in 5 seconds. Calculate the acceleration during each phase. State whether each value represents acceleration or deceleration, and explain how you know from the sign.**	Note: The average speed and average velocity have very different magnitudes because the rider doubled back — direction mattered. (c) Phase 1 (accelerating): u = 0 m/s, v = 25 m/s, t = 10 s. $a = (25 - 0) \div 10 = +2.5$ m/s². Positive value → acceleration (velocity increasing). Phase 2 (braking): u = 25 m/s, v = 0 m/s, t = 5 s. $a = (0 - 25) \div 5 = -5$ m/s². Negative value → deceleration (velocity decreasing). The negative sign shows the acceleration acts opposite to the direction of motion.

Section 3: Distance–Time and Velocity–Time Graphs	
Graphs are a powerful mathematical tool for representing and interpreting motion. **(a) For a distance–time (d–t) graph, describe what the following features represent about motion:** **(i) a straight line with a positive gradient** **(ii) a horizontal line** **(iii) a steeper straight line compared to a less steep one.** **(b) For a velocity–time (v–t)**	(a)(i) A straight line with a positive gradient on a d–t graph means the object is moving at constant speed — distance increases steadily with time. The gradient equals the speed. (a)(ii) A horizontal line means the object is stationary — distance is not changing, so speed = 0. (a)(iii) A steeper line means a greater gradient, which means a higher speed. The steeper walker in the Timer Race Puzzle covered the same distance in less time. (b)(i) A straight line with a positive gradient on a v–t graph means uniform acceleration — velocity increases at a constant rate. The gradient equals the acceleration. (b)(ii) A horizontal line means constant velocity (zero acceleration) — the object moves at a steady speed in one direction. (b)(iii) The area under a v–t graph equals the distance (or displacement, if direction is considered) travelled during that time interval. (c) [Graph described]: The v–t graph has time (s) on the x-axis (0 to 20 s) and velocity (m/s) on the y-axis (0 to 6 m/s). Three clearly labelled points: (0, 0), (5, 6), (15, 6), (20, 0). The graph shows a rising straight line from 0–5 s, a horizontal line from 5–15 s, and a falling straight line from 15–20 s.

graph, describe what the following features represent: **(i) a straight line with a positive gradient** **(ii) a horizontal line** **(iii) the area under the graph.** **(c) Eliud Kipchoge completes a training run described by the following data. Draw a velocity–time graph for this run, then use it to calculate the total distance covered.** **• 0–5 s: accelerates uniformly from 0 m/s to 6 m/s** **• 5–15 s: runs at a constant 6 m/s** **• 15–20 s: decelerates uniformly from 6 m/s to 0 m/s** **Label your axes fully. Show how you calculate the total distance from the graph (split the area into shapes and calculate each).**	Total distance = area under graph, split into three shapes: – Triangle (0–5 s): area = ½ × base × height = ½ × 5 × 6 = 15 m – Rectangle (5–15 s): area = length × width = 10 × 6 = 60 m – Triangle (15–20 s): area = ½ × 5 × 6 = 15 m Total distance = 15 + 60 + 15 = 90 m. Eliud covered 90 m in 20 seconds during this training interval.

Section 4: Using Equations of Linear Motion to Predict

The equations of uniformly accelerated motion allow us to predict where an object will be and how fast it will be moving — without having to run the experiment again. **(a) State the three equations of uniformly accelerated motion. Define every symbol used.** **(b) A maize delivery lorry travelling along the Nakuru–**	(a) The three equations of uniformly accelerated motion (where u = initial velocity, v = final velocity, a = acceleration, t = time, and s = displacement) are: 1. $v = u + at$ 2. $s = ut + \frac{1}{2}at^2$ 3. $v^2 = u^2 + 2as$ (b) Given: u = 20 m/s, a = −4 m/s² (deceleration), v = 0 m/s (stops). (i) Using $v = u + at$: $0 = 20 + (-4)t \rightarrow 4t = 20 \rightarrow t = 5$ s. The lorry takes 5 seconds to stop. (ii) Using $v^2 = u^2 + 2as$: $0^2 = 20^2 + 2(-4)s \rightarrow 0 = 400 - 8s \rightarrow 8s = 400 \rightarrow s = 50$ m. The lorry travels 50 m before stopping. (iii) The lorry needs 50 m to stop, but the road block is only 40 m away. Since 50 m > 40 m, the lorry does NOT stop in time and will

Nairobi highway is moving at 20 m/s when the driver sees a road block ahead and applies the brakes. The lorry decelerates uniformly at 4 m/s². **(i) How long does it take the lorry to stop?** **(ii) How far does the lorry travel before stopping?** **(iii) If the road block is only 40 m away when the driver brakes, does the lorry stop in time? Show your working and explain your answer clearly.** **(c) Connect this back to the Timer Race Puzzle: explain how equations of motion give us a 'richer mathematical language' that distance and speed alone cannot provide. What extra information do the equations give us that helps solve puzzles like the one in the phenomenon?**	collide with the road block. This has real safety implications — knowing stopping distances mathematically can save lives. (c) In the Timer Race Puzzle, knowing only distance and speed left us confused: how could the same numbers describe such different situations? The equations of motion include displacement (a vector), velocity (a vector), acceleration (which can be positive or negative to show direction of change), and time. By including direction through signs (positive/negative) and by distinguishing displacement from distance, these equations fully describe WHERE an object ends up, HOW FAST it is going, and in WHICH DIRECTION — not just how far it has travelled. This is the richer mathematical language the puzzle was calling for.

Section 5: Synthesis — Answering the Driving Question	
You have now used all the mathematical tools from this unit. In this final section, write a full, connected explanation that directly answers the driving question: **'How can mathematics help us fully describe, graph, and predict the motion of objects moving in a straight line — and why does direction always matter?'** **Your response must:** **– Be written in continuous**	At the start of this unit, the Timer Race Puzzle left us confused: how could walkers who covered the same distance end up in completely different places? Mathematics — specifically the mathematics of linear motion — gives us the precise language to resolve that confusion. The first key insight is the difference between scalar and vector quantities. Distance and speed are scalars: they record how much, but not which way. Displacement and velocity are vectors: they record both how much and in which direction. The zigzag walker and the straight-line walker had the same distance (a scalar) but very different displacements (vectors), because direction was ignored by distance but captured by displacement. That single distinction dissolves the puzzle entirely. Graphs deepen our understanding further. A distance–time graph shows us whether an object is moving or stationary and how fast it travels, through the gradient of the line. A velocity–time graph is even more powerful: its gradient tells us acceleration, and the area beneath it tells us displacement. By drawing and reading these graphs, we can interpret motion visually — without needing to be present at the event. Finally, the equations of uniformly accelerated motion ($v = u + at$; $s = ut + \frac{1}{2}at^2$; $v^2 = u^2 + 2as$) allow us to predict motion precisely.

prose (full paragraphs), not bullet points. **– Reference at least THREE mathematical tools or concepts from the unit (e.g., displacement, velocity–time graphs, equations of motion, acceleration, scalar vs. vector).** **– Return explicitly to the Timer Race Puzzle — explain how you would now resolve the 'puzzle' that confused you at the start of the unit.** **– Include at least ONE real Kenyan example of your own choice (not one already used in Sections 1–4) to illustrate your explanation.** **– Be between 200 and 350 words.**	Consider a fisherman at the Kisumu port accelerating his motorboat from rest at 1.5 m/s² for 8 seconds. Without running the experiment again, we can calculate that his final velocity is 12 m/s and that he has travelled 48 m — in a specific direction out into Lake Victoria. That predictive power is only possible because we use vectors, not scalars. Direction always matters because the physical world is not one-dimensional: objects move towards or away from destinations, they speed up or slow down, and the sign of their acceleration determines whether they are gaining or losing speed. Mathematics — through vectors, graphs, and equations — turns a confusing puzzle into a fully described, predictable, and understandable picture of motion.

RUBRIC			
Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Conceptual Understanding: Scalar vs. Vector	Clearly and correctly defines all four quantities (distance, displacement, speed, velocity) with examples. Consistently and accurately distinguishes scalar from vector throughout all sections. Uses direction (including positive/negative signs) correctly in every calculation and explanation.	Correctly defines most quantities and distinguishes scalar from vector in most instances. Direction is included in answers but may be stated informally (e.g., 'to the west') rather than mathematically (e.g., negative sign with justification) in one or two places.	Shows partial understanding — may confuse distance with displacement or speed with velocity in at least one section. Direction is often omitted from vector answers. The distinction between scalar and vector is stated but not consistently applied.
Mathematical Accuracy: Calculations and Use of Formulae	All formulae are correctly stated with symbols defined and SI units given. All calculations are fully worked (step by step), arithmetically correct, and carry the correct units throughout. Rounding is appropriate and explicitly stated where used.	Correct formulae are used in most cases. Working is shown for most calculations and is largely correct, with at most one arithmetic or unit error that does not propagate through the rest of the answer.	Some correct formulae are used but others are missing or incorrectly recalled. Working is partial or missing in several places. Multiple arithmetic or unit errors are present, making some answers unreliable.

Graphical Representation and Interpretation	Velocity–time graph (Section 3) is accurately drawn with fully labelled axes (quantity and unit), at least three correctly plotted and labelled points, and straight-line segments drawn accurately. Area is correctly split into named geometric shapes and each area is calculated correctly with appropriate units.	Graph is drawn with labelled axes and mostly correct plotting. One minor error in plotting or labelling is present. Area calculation identifies the correct shapes and arrives at the correct or nearly correct total distance, with working shown.	Graph is attempted but has significant errors — axes may be unlabelled or incorrectly scaled, points are missing or incorrectly placed, or the graph is a curve rather than straight-line segments. Area calculation is attempted but incorrect or missing.
Equations of Motion: Application and Prediction	All three equations are correctly stated and defined in Section 4(a). Correct equation(s) are selected and applied in each part of 4(b). The stopping-distance comparison in 4(b)(iii) is explicitly resolved with a clear mathematical conclusion. The conceptual explanation in 4(c) accurately articulates what the equations add beyond scalar descriptions.	Two or three equations are correctly stated and mostly applied correctly. The stopping-distance conclusion in 4(b)(iii) is reached, though the comparison to 40 m may lack explicit mathematical wording. The explanation in 4(c) connects equations to direction but may be brief.	Only one or two equations are correctly recalled. Applications in 4(b) contain errors or missing steps. The conclusion in 4(b)(iii) is missing or incorrect. The explanation in 4(c) is vague or does not connect equations to the concept of direction or vectors.
Synthesis and Communication: Answering the Driving Question	Section 5 response is written in fluent, continuous prose within the word range (200–350 words). It references at least three distinct mathematical tools from the unit with accurate explanation. The Timer Race Puzzle is explicitly and accurately resolved. At least one original, plausible Kenyan example is included and correctly used to support the explanation. Mathematical vocabulary (scalar, vector, displacement, velocity, acceleration, gradient, area under graph) is used accurately throughout.	Response is in prose and references at least two mathematical tools clearly. The Timer Race Puzzle is mentioned and partially resolved. A Kenyan example is included but may be briefly explained. Mathematical vocabulary is mostly correct with one or two imprecise uses.	Response uses bullet points or is very brief (under 150 words). Fewer than two mathematical tools are referenced, or they are described inaccurately. The Timer Race Puzzle is not meaningfully addressed. A Kenyan example is absent or irrelevant. Mathematical vocabulary is limited or misused.